

Java 8 Method and Constructor Reference - Exercises (Basic to Advanced)

1. Create a functional interface named Printable with a method print(). Use method reference to refer to a static method that prints a message.
2. Define a class with an instance method show(). Use method reference to refer to this method via an object.
3. Use method reference to sort a list of strings in alphabetical order using Collections.sort().
4. Use method reference to convert a list of strings to uppercase using List.forEach().
5. Create a method that returns the square of a number. Use method reference to refer to this method using Function.
6. Define a functional interface and use constructor reference to create objects of a class.
7. Sort a list of objects (e.g., students by age) using a method reference to an instance method of an arbitrary object.
8. Use BiFunction to refer to a static method that adds two numbers.
9. Use method reference to filter a list of integers and print only even numbers.
10. Create a reference to a method that checks whether a string is empty and use it in a Predicate.
11. Use method reference inside a Stream API pipeline to transform and filter data.
12. Create a list of employees and use method references to map names and sort by salary.
13. Use a constructor reference with a Supplier to instantiate objects.
14. Combine method references with Optional API to perform safe operations on values.
15. Write a program using a method reference to a generic method that compares two values and returns the larger one.